Quantum Field Theory and the Standard Model – Matthew D. Schwartz

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2026

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Part I

Field Theory

1 Preliminary Details

1.1 $\hbar = c = 1$

Physicists like to set $\hbar = c = 1$ to 1) simplify equations, 2) reveal some symmetries. This makes all quantities some power of mass.

Note that:

$$c = 1 : [\text{Length}] = [\text{Time}]$$
$$\hbar = 1 : [\text{Time}]^{-1} = [\text{Energy}] = [\text{Mass}]$$

- Quantities with positive mass dimensions are "Energies."
- Quantities with negative mass dimensions are "Lengths."

Then further note:

$$[dx] = [x] = [t] = -1$$
$$[\partial_\mu] = [p_\mu] = 1$$

You'll get used to it.

1.2 Lorentz Transformations

Lorentz transformations are composed of rotations + boosts. They're generated by the **Lorentz group.** They preserve the Minkowski inner product:

$$\boxed{\Lambda^T g \Lambda = g}$$

i.e. $V_\mu W^\mu = V_0 W_0 - V_1 W_1 - V_2 W_2 - V_3 W_3$ is invariant.

- Note that we use $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$ here.
- And that $g_{\mu\nu} = \eta_{\mu\nu}$ outside of exotic quantum gravity stuff.

$$\text{Timelike} : V^\mu V_\mu > 0$$
$$\text{Spacelike} : V^\mu V_\mu < 0$$
$$\text{Lightlike/Null} : V^\mu V_\mu = 0$$

2 Historic Background

2.1 Blackbody Radiation

Blackbody: an object at some temperature (internal structure is irrelevant)

- A box with size L has EM waves s.t.

$$\lambda = L/n \ , \quad v = f\lambda$$
$$\omega_n = \frac{2\pi}{L}|\vec{n}|c$$

- **Classical equipartition theorem:** light is emitted equally in all modes. The intensity is proportional to the differential volume of phase space, thus by dimensional analysis,

$$I(\omega) \equiv \frac{1}{V}\frac{d}{d\omega}E(\omega) \ \longrightarrow \ I(\omega) = \left[c^{-3}\omega^2 k_B T\right]$$

We expect $I(\omega)$ to grow with angular frequency (ω), but this experimentally the graph is a Maxwell-Boltzmann distribution. Called the "ultraviolet catastrophe."

Planck's solution: Energy in each EM mode is quantized.

$$\boxed{E_n = \hbar\omega_n}$$

- Interpretation: light is made up of massless particles ("photons")
- If we calculate $I(\omega)$ using this, the catastrophe is averted.
- The derivation (see book) uses equilibrium stat mech – so light must "equilibriate," i.e. particles must be created & destroyed. This is the domain of QFT.

$$I(\omega) = \frac{1}{V}\frac{dE(\omega)}{d\omega} = \frac{\hbar}{\pi^2}\frac{\omega^3}{e^{\hbar\omega\beta} - 1}$$

2.2 Second Quantization

2.2.1 Einstein Coefficients

Assume an atom with energy levels E_1, E_2 such that $E_2 - E_1 = \hbar\omega$. Further assume n_2 of the E_2 atoms and n_1 of the E_1 atoms.

$$\begin{aligned} A &: \text{Coefficient of spontaneous emission} \\ B &: \text{Coefficient of stimulated emission} \\ B' &: \text{Coefficient of absorption} \end{aligned}$$

A is the probability for a E_2 atom to become a E_1 atom by emitting a $\hbar\omega$ photon. B is the probability for an incoming $\hbar\omega$ photon to induce $E_2 \to E_1$. B' is the probability for an incoming photon to induce $E_1 \to E_2$.

We can estimate the differential dn_2 based on the above. Further, since particle number is conserved in this system,

$$\begin{aligned} dn_2 &\approx -\left[A + BI(\omega)\right] n_2 \\ dn_1 = -dn_2 &= -\left[A + BI(\omega)\right] n_2 + B'I(\omega)n_1 \end{aligned}$$

Using equilibrium stat mech, Einstein was able to predict the relation between A, B, and B'.

2.2.2 Dirac

Recall the quantum harmonic oscillator with creation/destruction operators $a^\dagger$, a.

$$\begin{aligned} [a, a^\dagger] &= 1 \\ \hat{N} \equiv a^\dagger a\ , &\quad \hat{N}\left|n\right\rangle = n\left|n\right\rangle \\ a^\dagger\left|n\right\rangle &= \sqrt{n+1}\left|n+1\right\rangle \\ a\left|n\right\rangle &= \sqrt{n}\left|n-1\right\rangle \end{aligned}$$

(Be able to derive above) Now we use Fermi's golden rule to calculate the transition rates.

$$\Gamma \sim |\mathcal{M}|^2 \delta(E_f - E_i) \ , \quad \mathcal{M} = \langle f | H_{\text{int}} | i \rangle$$

H_{int} must have creation/destruction operators to create photons. It must also be Hermitian. Thus $H_{\text{int}} = H_I^\dagger a^\dagger + H_I a$. For the $2 \to 1$ transition, $|i\rangle = |\text{atom}_2\rangle \otimes |n_\omega\rangle$ and $|f\rangle = |\text{atom}_1\rangle \otimes |n_\omega + 1\rangle$ where n_ω (number of photons) has increased by one.

$$\begin{aligned}
\mathcal{M}_{2\to1} &= \langle f | H_{\text{int}} | i \rangle \\
&= \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle \langle n_\omega + 1 | a^\dagger | n_\omega \rangle + \langle \text{atom}_1 | H_I | \text{atom}_2 \rangle \langle n_\omega + 1 | a | n_\omega \rangle \\
&= \mathcal{M}_0^\dagger \sqrt{n_\omega + 1} \ , \quad \mathcal{M}_0^\dagger \equiv \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle
\end{aligned}$$

The $1 \to 2$ transition is very similar.

$$\mathcal{M}_{1\to2} = \mathcal{M}_0 \sqrt{n_\omega} \ , \quad \mathcal{M}_0 \equiv \langle \text{atom}_2 | H_I | \text{atom}_1 \rangle$$

Thus n_2 and n_1 change as

$$\begin{aligned}
dn_2 = -dn_1 &= -|\mathcal{M}_{2\to1}|^2 n_2 + |\mathcal{M}_{1\to2}|^2 n_1 \\
&= -|\mathcal{M}_0|^2 (n_\omega + 1) n_2 + |\mathcal{M}_0|^2 (n_\omega) n_1
\end{aligned}$$

Now we start the heavy calculations.

1. Energy is quantized by $\Delta = \hbar\omega = \hbar \frac{2\pi}{L} |\vec{n}|$ (with $c = 1$)

$$E(\omega) = \int^\omega (\hbar\omega) n_\omega \ d^3\vec{n} = (4\pi) \hbar L^3 \int_0^\omega \frac{d\omega}{(2\pi)^3} \omega^3 n_\omega$$

2. Multiply above by 2 for the 2 polarizations of light.

3. Intensity is

$$I(\omega) = \frac{1}{L^3} \frac{dE}{d\omega} = \frac{\hbar\omega^3}{\pi^2} n_\omega$$

So now we have

$$\boxed{dn_2 = -dn_1 = -|\mathcal{M}_0|^2 \left[1 + \frac{\pi^2}{\hbar\omega^3} I(\omega) \right] + |\mathcal{M}_0|^2 \left[\frac{\pi^2}{\hbar\omega^3} I(\omega) \right] n_1}$$

By comparison with the previous section, we read off the Einstein coefficients:

$$\boxed{B' = B \ , \quad \frac{A}{B} = \frac{\hbar}{\pi^2} \omega^3}$$

2.2.3 Key Ideas

Technically only one "quantization" is going on.

First quantization: Interpret modes as having quantized energy $E = \hbar\omega$
Use Hilbert space $\mathcal{H}_n$ for n-particle state
Second quantization: Quantize each mode as a harmonic oscillator
Use Fock space $\mathcal{F} = \oplus_n \mathcal{H}_n$ over all $0 \dots n$-particle states

2.3 Harmonic Oscillators

2.3.1 Lorentz Invariance and Special Relativity

A simple Lorentz-invariant equation is

$$\Box\phi = (\partial_t^2 - \nabla^2)\phi = 0$$

An example solution is $\phi(x) = a_p(t)e^{i\vec{p}\cdot\vec{x}}$ with $(\partial_t^2 + \vec{p}\cdot\vec{p})a_p(t) = 0$. This is very similar to the harmonic oscillator equation $m\ddot{x} + kx = 0$.

The **general solution** sums over all p modes.

$$\phi(\vec{x},t) = \int \frac{d^3p}{(2\pi)^3}\left[a_p(t)e^{i\vec{p}\cdot\vec{x}} + a_p^*(t)e^{-i\vec{p}\cdot\vec{x}}\right]$$

$$\text{Or: } \phi(\vec{x},t) = \int \frac{d^3p}{(2\pi)^3}\left(a_p e^{-ipx} + a_p^* e^{ipx}\right), \quad p_\mu \equiv (\omega_p, \vec{p}) \text{ and } px = p_\mu x^\mu$$

- This shows the decomposition of a scalar field into plane waves.
- Some vector fields can be dealt with this as well. Electromagnetic waves, for example, obey $\Box A_\nu = 0$ in each of the A_ν of the vector potential, given the Lorenz gauge $\partial_\mu A_\mu = 0$.

2.3.2 In Quantum Field Theory

We generalize the harmonic oscillator of quantum mechanics, $[a, a^\dagger] = 1$, to $[a_k, a_p^\dagger] = (2\pi)^3\delta^3(\vec{p} - \vec{k})$ with factors for convention.

As expected (with factors for convention),

$$a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |\vec{p}\rangle$$
$$\langle 0| a_p = \frac{1}{\sqrt{2\omega_p}} \langle\vec{p}|$$

Quantum fields are momentum integrals over these creation/annihilation operators for multiple oscillators.

$$\boxed{\phi_0(\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right), \quad px \equiv p_\mu x^\mu}$$

- By acting on $|0\rangle$ and projecting onto the momentum space, we see that
$$\langle\vec{p}|\phi_0(x)|0\rangle = e^{-ipx}$$
- I.e. the action of $\phi_0(x)$ is to create particles at position $\vec{x}$.

2.3.3 Time Dependence

In QFT, we often use the Heisenberg picture: time dependence is encoded in the operators.

$$a_p(t) = e^{-i\omega_p t} a_p$$
$$a_p^\dagger(t) = e^{i\omega_p t} a_p^\dagger$$

Thus we could **define** quantum scalar fields as

$$\boxed{\phi_0(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{-ipx} + a_p^\dagger e^{ipx} \right)}$$